

Experiment 1

Single phase half wave uncontrolled rectifier

1. Abstract

In the experiment we will construct the single phase half wave rectifier circuit and investigate its output with R, RL, and RC loads. The purpose of this circuit is to provide a DC output from an AC source.

2. Introduction and Theory

This circuit uses one diode to block the voltage applied in reverse bias to it, i.e. it will allow only one half of the AC wave to pass on to the load, whether it's the positive or the negative half depends on the diodes orientation in the circuit, so we could use this circuit to give us a DC voltage output.

3. Used equipments

1. Diode
2. AC power supply
3. Circuit breakers
4. R, R-L and R-C loads
5. Oscilloscope
6. Digital Multimeter
7. Connecting wires

4. Circuit diagram

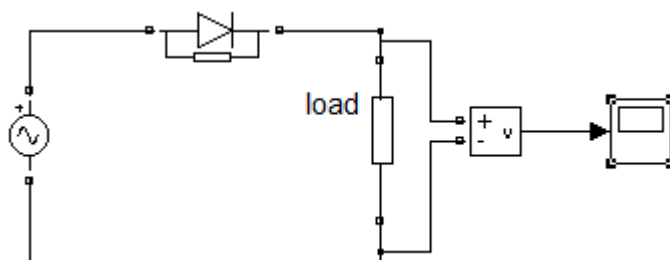


Figure 1.1

5. Experimental methods

- **Resistive load**

- 1) Construct the circuit diagram with resistive load shown in figure (1-1).
- 2) Check your circuit connection with the help of your supervisor.
- 3) Turn on the power supply.
- 4) Record the output signal in table (1-1).
- 5) Turn off the power supply.
- 6) Calculate $V_R, V_o, I_R, I_o, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (1-2).
- 7) Compare your calculations with the measured values.
- 8) Draw the output wave forms.

Table 1.1 measured values

Load	V_i	V_o	V_R	I_o	I_R	fi	fo
R							
RL							
RL with freewheeling diode							
RC							

Table 1.2 calculated values

Load	V_i	V_o	V_R	I_o	I_R	P_{dc}	P_{ac}	FF	RF	η
R										
RL										
RL with freewheeling diode										

RC										
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- **Resistive and inductive load**
 - 1) Construct the circuit diagram with resistive and inductive load.
 - 2) Repeat steps (2-8).

- **Resistive and inductive load with freewheeling diode**
 - 1) Construct the circuit diagram with resistive and inductive with freewheeling diode load.
 - 2) Repeat steps (2-8).

- **Resistive and capacitive load**
 - 1) Construct the circuit diagram with resistive and capacitive load.
 - 2) Repeat steps (2-8).

6. Conclusions

Experiment 2

Single phase full wave uncontrolled rectifier

1. Abstract

In the experiment we will construct the single phase full-wave rectifier circuit and investigate its output with R, RL, and RC loads.

2. Introduction and Theory

This circuit is an AC to DC converter; there are two major kinds of this converter: Bridge circuit with four diodes and the center tapped uses only two diodes, but it needs a center tapped transformer.

3. Used equipments

- 1- Diodes.
- 2- AC power supply.
- 3- Circuit breakers.
- 4- R, L and C loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.
- 8- Center tapped transformer.

4. Experimental methods

A) Center tapped Circuit

Circuit Diagram

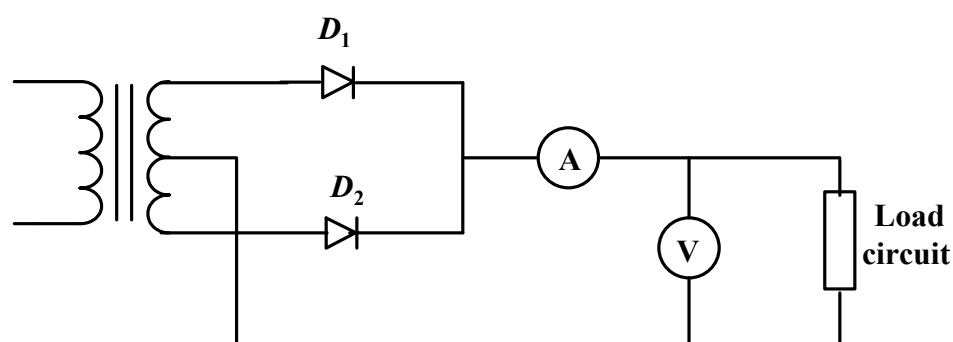


Figure 2.1

• Resistive load

- 1) Construct the circuit diagram with resistive load shown in figure (2-1).
- 2) Check your circuit connection with the help of your supervisor.
- 3) Turn on the power supply.

- 4) Record the output signal in table (2-1).
- 5) Turn off the power supply.
- 6) Calculate $V_R, V_o, I_R, I_o, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (2-2).
- 7) Compare your calculations with the measured values.
- 8) Draw the output wave forms.

Table 2.1 measured values

Load	V_i	V_o	V_R	I_o	I_R	P_{dc}	P_{ac}	FF	RF	η
R										
RL										
RL with freewheeling diode										
RC										

Table 2.2 calculated values

Load	V_i	V_o	V_R	I_o	I_R	f_i	f_o
R							
RL							
RL with freewheeling diode							
RC							

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

B) Full bridge circuit

Circuit Diagrams

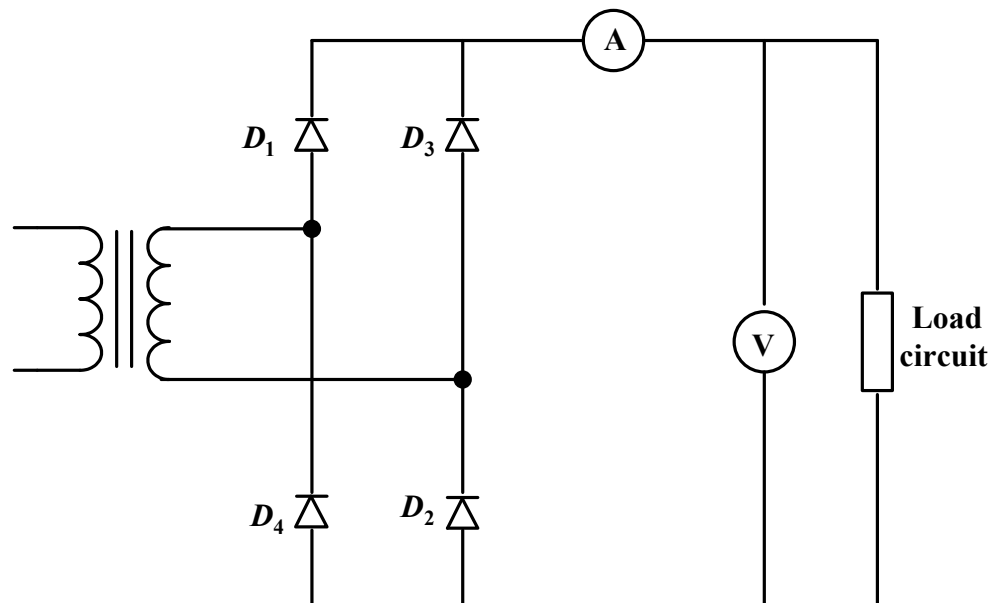


Figure 2.2

- **Resistive load**

- 1) Construct the circuit diagram with resistive load shown in figure (2-2).
- 2) Check your circuit connection with the help of your supervisor.
- 3) Turn on the power supply.
- 4) Record the output signal in table (2-3).
- 5) Turn off the power supply.
- 6) Calculate $V_R, V_0, I_R, I_0, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (2-4).
- 7) Compare your calculations with the measured values.
- 8) Draw the output wave forms.

Table (2-3) measured values

Load	V_i	V_o	V_R	I_o	I_R	f_i	f_o
R							
RL							
RL with freewheeling diode							
RC							

Table (2-4) calculated values

Load	V_i	V_o	V_R	I_o	I_R	P_{dc}	P_{ac}	FF	RF	η
R										
RL										
RL with freewheeling diode										
RC										

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- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

5. Conclusions

Experiment 3

Three phase half wave uncontrolled rectifier

1. Abstract

In this experiment we will investigate the operation of the three phase half wave rectifier with different load configurations, AC and DC voltages as well as waveforms and frequencies of the outputs are to be found.

2. Introduction and Theory

This circuit convert's three phase AC voltage to a DC voltage delivered to the load. It utilizes one diode for each phase, and only one phase will be connected to the load at anytime, which means that only one diode will be ON at anytime.

The anode of each diode is connected to each phase voltage source. The cathodes of all three diodes are connected together to form the positive load terminal. The negative terminal of the load is connected to the supply neutral.

3. Used equipments

- 1- Diodes.
- 2- AC power supply.
- 3- Circuit breakers.
- 4- R and L loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.

4. Circuit diagram

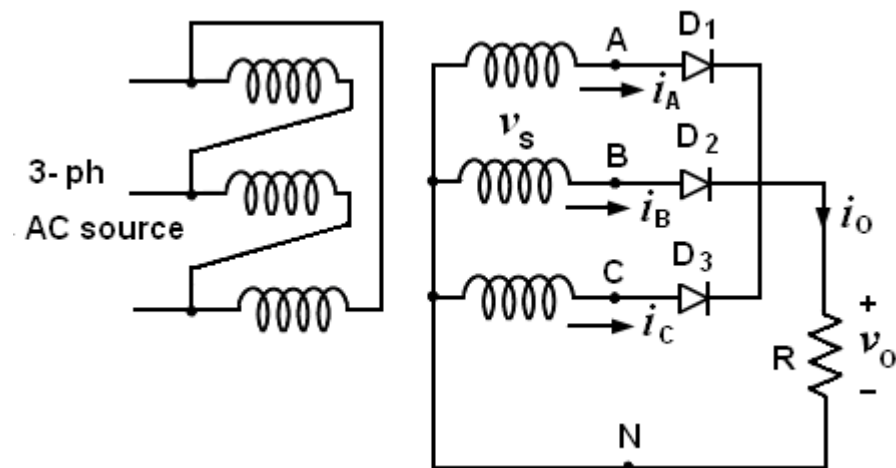


Figure 3.1

5. Experimental methods

- **Resistive load**

- 1) Construct the circuit diagram with resistive load shown in figure (3-1).
- 2) Check your circuit connection with the help of your supervisor.
- 3) Turn on the power supply.
- 4) Record the output signal in table (3-1).
- 5) Turn off the power supply.
- 6) Calculate $V_R, V_0, I_R, I_0, RF, P_{dc}, P_{ac}, \eta, \text{and } FF$, and record your calculation in table (3-2).
- 7) Compare your calculations with the measured values.
- 8) Draw the output wave forms.

Table 3.1 measured values

Load	V_i	V_o	V_R	I_o	I_R	f_i	f_o
R							
RL							
RL with freewheeling diode							
RC							

Table 3.2 calculated values

Load	V_i	V_o	V_R	I_o	I_R	P_{dc}	P_{ac}	FF	RF	η
R										
RL										
RL with freewheeling diode										
RC										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

6. Conclusions

Experiment 4

Three phase full wave uncontrolled rectifier

Abstract .1

In this experiment we will investigate the operation of the six diodes full wave converter with different load configurations, AC and DC voltages .as well as waveforms and frequencies of the outputs are to be found

Introduction and Theory .2

To understand the operation sequence of the diodes it is necessary for the diodes to be labeled as in the circuit below. The time interval during which a particular conduction mode will be effective can be ascertained from the following: For example the D1D2 conduction mode will occur when the voltage across all other diodes (i.e. V_{ba} , V_{ca} and V_{cb}) are negative. This implies that D1D2 conducts in the interval $0 \leq \omega t \leq \pi/3$. The diodes have been numbered such that the conduction sequence is D1 $\rightarrow$ D2 $\rightarrow$ D3 $\rightarrow$ D4 $\rightarrow$ D5 $\rightarrow$ D6 $\rightarrow$ D1. When a diode stops conduction its current is commutated to another diode in the same group (top or bottom). This way the sequence of conduction modes becomes: D6D1 $\rightarrow$ D1D2 $\rightarrow$ D2D3 $\rightarrow$ D3D4 $\rightarrow$ D4D5 $\rightarrow$ D5D6 $\rightarrow$ D6D1 and so on

Used equipments .3

- .Diodes -1
- .AC power supply -2
- .Circuit breakers -3
- .R, L and C loads -4
- .Oscilloscope -5
- .Digital Millimeter -6
- .Connecting wires -7

Circuit Diagram .4

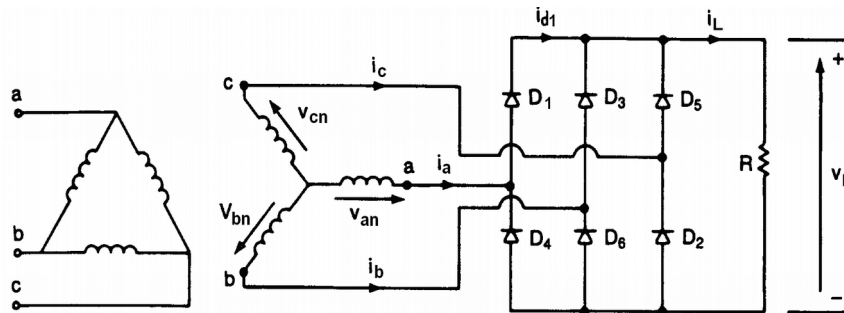


Figure 4.1

Experimental methods .5

- **Resistive load**

- 1) Construct the circuit diagram with resistive load shown in figure (4-1).
- 2) Check your circuit connection with the help of your supervisor.
- 3) Turn on the power supply.
- 4) Record the output signal in table (4-1).
- 5) Turn off the power supply.
- 6) Calculate V_R , V_o , I_R , I_o , RF , P_{dc} , P_{ac} , η , and FF , and record your calculation in table (4-2).
- 7) Compare your calculations with the measured values.
- 8) Draw the output wave forms.

Table 4.1 measured values

Load	V_i	V_o	V_R	I_o	I_R	f_i	f_o
R							
RL							
RL with freewheeling diode							
RC							

Table 4.2 calculated values

Load	V_i	V_o	V_R	I_o	I_R	P_{dc}	P_{ac}	FF	RF	η
R										
RL										
RL with freewheeling diode										
RC										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

Conclusions .6

Experiment 5

Triggering circuit

1. Abstract

In this experiment we will construct the triggering panel used as control panel for triggering the thyristors.

2. Technical specifications:

Supply voltage: +15 V, 0 V, and -15 V

Current consumption 100 mA

Synchronization voltage 1 V – 440 V $\pm$ 20%

Monitoring of the Synchronization voltage (phase failure, rotating field direction)

- Green LED, output to + 15 V for clockwise rotating field.
- Red LED, output to 0V for phase failure or counterclockwise field.

Test Voltage:

Control voltage U_{ST} 0V – 10 V.

Trigger pulses (main and subsequent pulse):

Number of output pulses 6 pulses

Voltage- time areas: 500 μ V_s

Pulse Train length: 2.5 ms (45°)

Pulses for pulse train: 50 pulses

3. Introduction and Theory

The six-pulse control unit is used to trigger Thyristors and IGBTs in controlled rectifier circuits, choppers and inverters.

The control unit is to be connected with a symmetric supply voltage $\pm 15\text{ V}$

A clockwise rotating field is to be connected to the synchronization input (4). The circuit itself produces a center point, for this reason, the N-conductor must not be connected.

* When the unit is used as a single-phase control unit, the N-conductor should be connected.

Trigger pulses are available at a six electrically isolated pulse outputs (5)-(10).

Depending on the control voltage (11) $U_{ST} = 0-10\text{ V}$, the phase of the trigger pulses is shifting from $180^\circ - 0^\circ$.

The inhibit input (13) can be used to switch off the trigger pulses, if the INH input is connected to earth, the trigger pulses are suppressed.

- Be careful that the polarities are always correct when connecting the electronic device to the control unit

4. Used equipments

1. Pulse generator circuit
2. DC power supplies
3. Circuit breakers
4. Oscilloscope
5. Digital Multimeter
6. Connecting wires

5. Circuit diagram

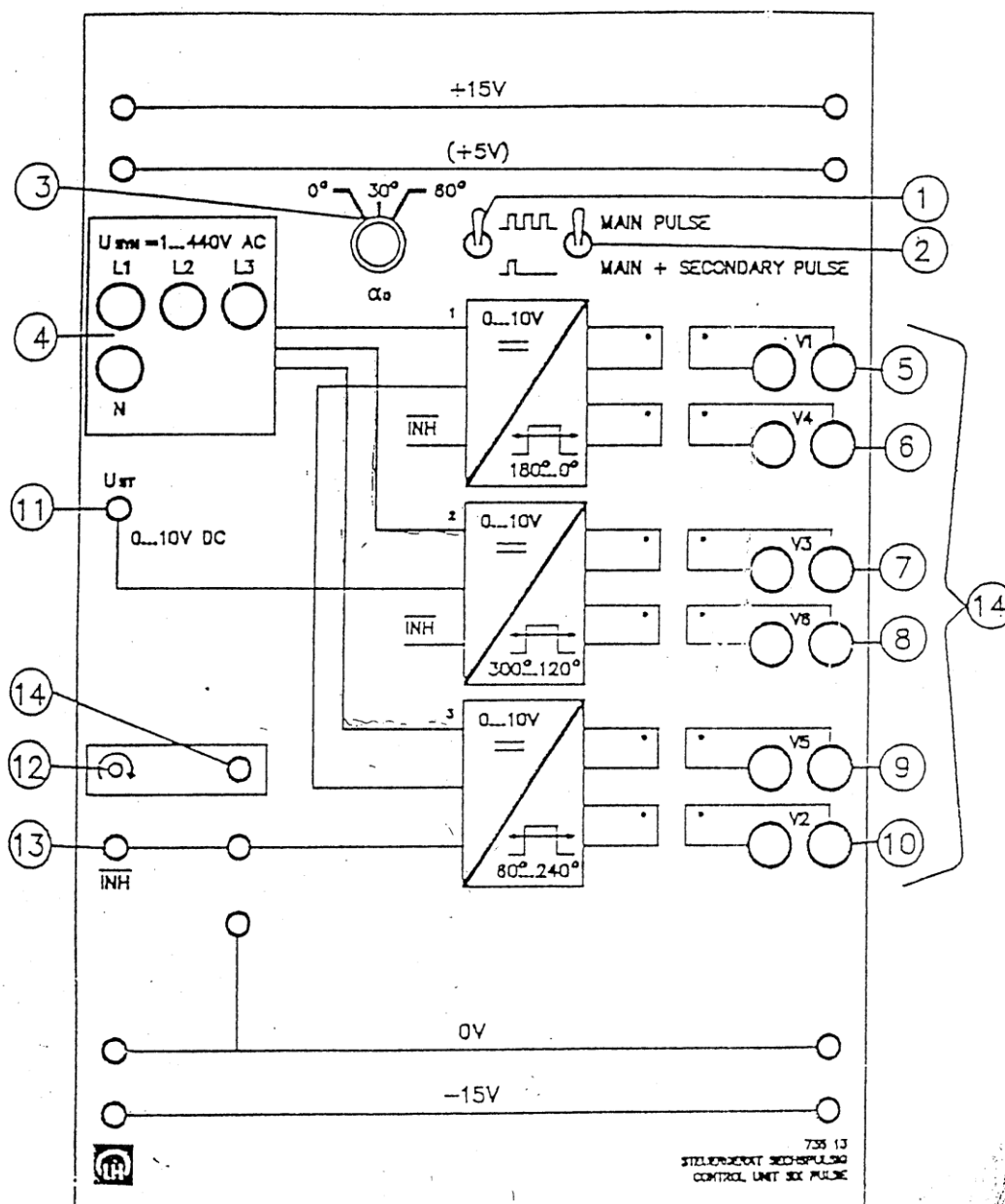


Figure 5.1

6. Experimental methods

We will use the following circuit elements provided to construct the circuit and perform the needed measurements; we will measure for different values of the reference DC voltage the trigger angle α and use the oscilloscope to find the value of α .

- 1- Construct the circuit diagram as shown in figure (5-1).
- 2- Check your circuit connection with the help of your supervisor.
- 3- Turn on the power supply.
- 4- Record the output signal in table (5-1).

Table (5-1) measured value

$U_{ST} \text{ V}$	0	2	5	7	10
α					

7. Conclusions

Single phase half wave controlled rectifier

1. Abstract

In this experiment we will construct the single phase half wave controlled rectifier circuit and investigate its output with R, RL, and RC loads.

An additional pulse generation circuit is needed for the trigger circuit of the thyristor.

2. Introduction and Theory

This type of rectifiers is called 'Controlled' because of the use of a Thyristor rather than a diode, the Thyristor has an extra terminal called the Gate, the thyristor, in addition to the necessary forward bias voltage, needs a pulse at the gate to start conduction, this process is called the firing or triggering of the thyristor. The firing angle (α) is controlled using the triggering circuit.

3. Used equipments

- 1-Thyristors.
- 2- AC power supply.
- 3- Circuit breakers.
- 4- R, L and C loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.
8. Pulse generator circuit.

4. Circuit Diagram

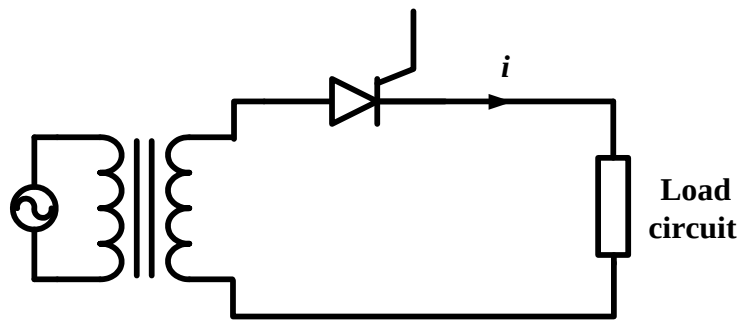


Figure 6.1

5. Experimental methods

- **Resistive load**

- 1- Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2- Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3- Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4- Check trigger signal using oscilloscope.
- 5- Turn off the power supply and construct the circuit diagram shown in figure (6-1).
- 6- Check your circuit connection with the help of your supervisor.
- 7- Turn on the power supply.
- 8- Record the output signal in table (6-1).
- 9- Turn off the power supply.
- 10- Calculate $V_R, V_0, I_R, I_0, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (8-1).
- 11- Compare your calculations with the measured values.
- 12- Draw the output wave forms.

Table 6.1 measured values for R load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					

150°					
180°					

Table 6.2 calculated values for R load

α	V_{in}	V_0	V_R	I_0	I_R	Pdc	pac	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

Table 6.3 measured values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 6.4 calculated values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R	Pdc	pac	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

• **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

Table 6.5 measured values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 6.6 calculated values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R	Pdc	pac	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

Table 6.7 measured values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 6.8 calculated values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R	Pdc	pac	FF	RF	η

0°										
30°										
60°										
90°										
150°										
180°										

6. Conclusions

Experiment 6

Single phase full wave controlled rectifier

1. Abstract

In the experiment we will construct the single phase full-wave controlled rectifier circuit and investigate its output with different loads.

2. Introduction and Theory

This circuit is an AC to DC converter; there are two major kinds of this converter: Bridge circuit with four thyristors and the center tapped uses only two thyristors, but it needs a center tapped transformer.

3. Used equipments

- 1- Thyristors.
- 2- AC power supply.
- 3- Circuit breakers.
- 4- R, L and C loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.
- 8- Center tapped transformer.
- 9- Firing Circuit or Pulse Generator.

4. Experimental methods

A) Center tapped Circuit

Circuit Diagrams

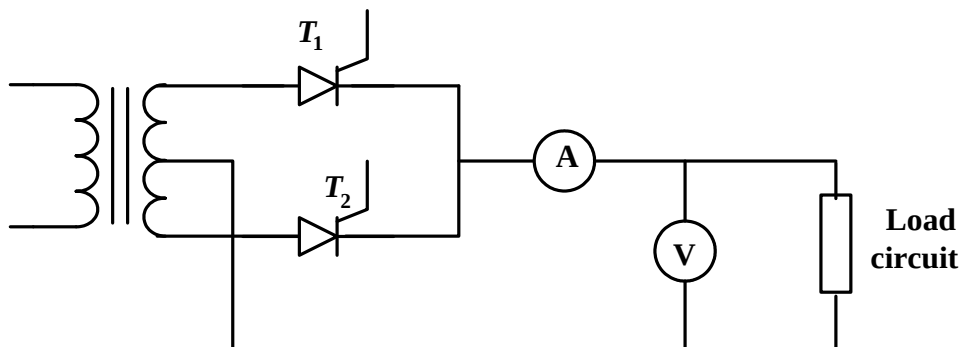


Figure 7.1

- **Resistive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (7-1).
- 6) Check your circuit connection with the help of your supervisor.
- 7) Turn on the power supply.
- 8) Record the output signal in table (7-1).
- 9) Turn off the power supply.
- 10) Calculate $V_R, V_0, I_R, I_0, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (7-1).
- 11) Compare your calculations with the measured values.
- 12) Draw the output wave forms.

Table 7.1 measured values for R load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.2 calculated values for R load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

Table 7.3 measured values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.4 calculated values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

Table 7.5 measured values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.6 calculated values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

Table 7.7 measured values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.8 calculated values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

B) Full bridge circuit

Circuit Diagrams

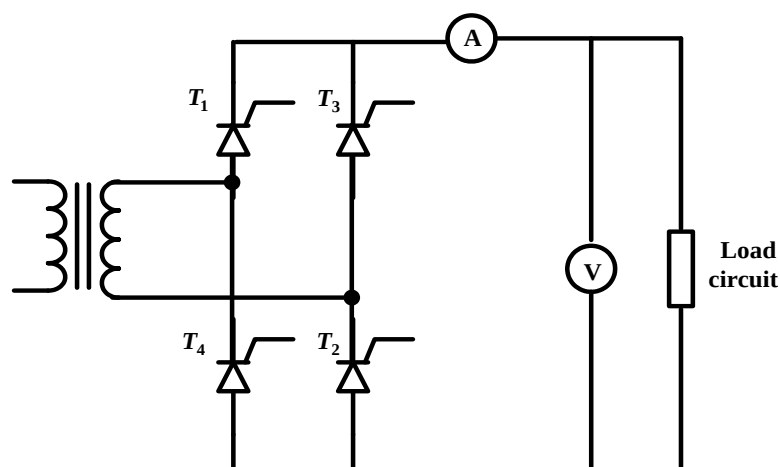


Figure 7.2

- **Resistive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (7-2).
- 6) Check your circuit connection with the help of your supervisor.
- 7) Turn on the power supply.
- 8) Record the output signal in table (7-9).
- 9) Turn off the power supply.
- 10) Calculate $V_R, V_0, I_R, I_0, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (7-10).
- 11) Compare your calculations with the measured values.
- 12) Draw the output wave forms.

Table (7-9) measured values for R load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table (7-10) calculated values for R load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

Table 7.11 measured values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.12 calculated values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive with freewheeling diode load**
 - 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
 - 2) Repeat steps (2-8).

Table 7.13 measured values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.14 calculated values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

Table 7.18 measured values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 7.16 calculated values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

5. Conclusions

Experiment 7

Three phase half wave controlled rectifier

1. Abstract

In this experiment we will build a construction of the 3ph half wave controlled rectifiers. We will do the experiment with different load configurations.

2. Introduction and Theory

The half wave circuit has 3 thyristors and 3 modes of operation, where each mode has one active thyristor.

3. Used equipments

- 1- Thyristors.
- 2- 3 phase AC power supply.
- 3- Circuit breakers.
- 4- R, and L loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.
- 8- Firing Circuit or Pulse Generator.

4. Circuit Diagrams

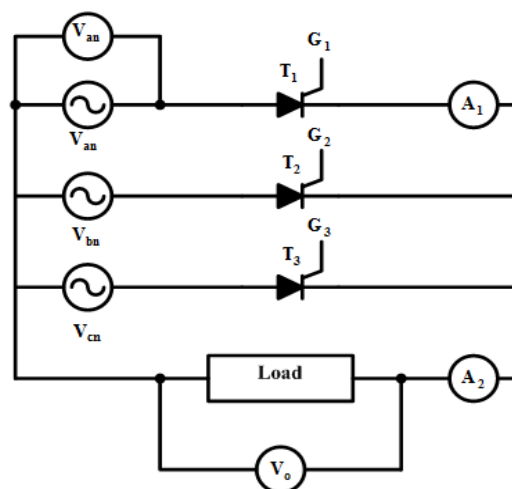


Figure 8.1

5. Experimental Methods

- **Resistive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (8-1).
- 6) Check your circuit connection with the help of your supervisor.
- 7) Turn on the power supply.
- 8) Record the output signal in table (8-1).
- 9) Turn off the power supply.
- 10) Calculate $V_R, V_0, I_R, I_0, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your calculation in table (8-1).
- 11) Compare your calculations with the measured values.
- 12) Draw the output wave forms.

Table 8.1 measured values for R load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 8.2 calculated values for R load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

Table 8.3 measured values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 8.4 calculated values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

Table 8.5 measured values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 8.6 calculated values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

Table 8.7 measured values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 8.8 calculated values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

6. Conclusions

Three phase full wave controlled rectifier

1. Abstract

In this experiment we will build a construction of the 3ph full wave controlled rectifiers, we will use the six thyristors bridge configuration for the full wave rectifiers. We will do the experiment with different load configurations.

2. Introduction and Theory

Like the Uncontrolled 3ph Bridge the load is connected to the circuit in such a way that it will have only line to line voltages across it while the full wave rectifier outputs line voltages, both phase and line voltages differ in phase by 120° .

The full wave circuit has 6 modes of operation, that is, we will have 6 different combinations of active thyristors and they are as follows: T_6T_1 , T_1T_2 , T_2T_3 , T_3T_4 , T_4T_5 , and T_5T_6 . We can see that we can have only 2 thyristors active at any instance.

As expected from a 3ph rectifier this circuit will have relatively high efficiency in AC to DC conversion.

3. Used equipments

- 1- Thyristors and diodes.
- 2- 3 phase AC power supply.
- 3- Circuit breakers.
- 4- R, and L loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.
- 8- Firing Circuit or Pulse Generator.

4. Circuit Diagrams

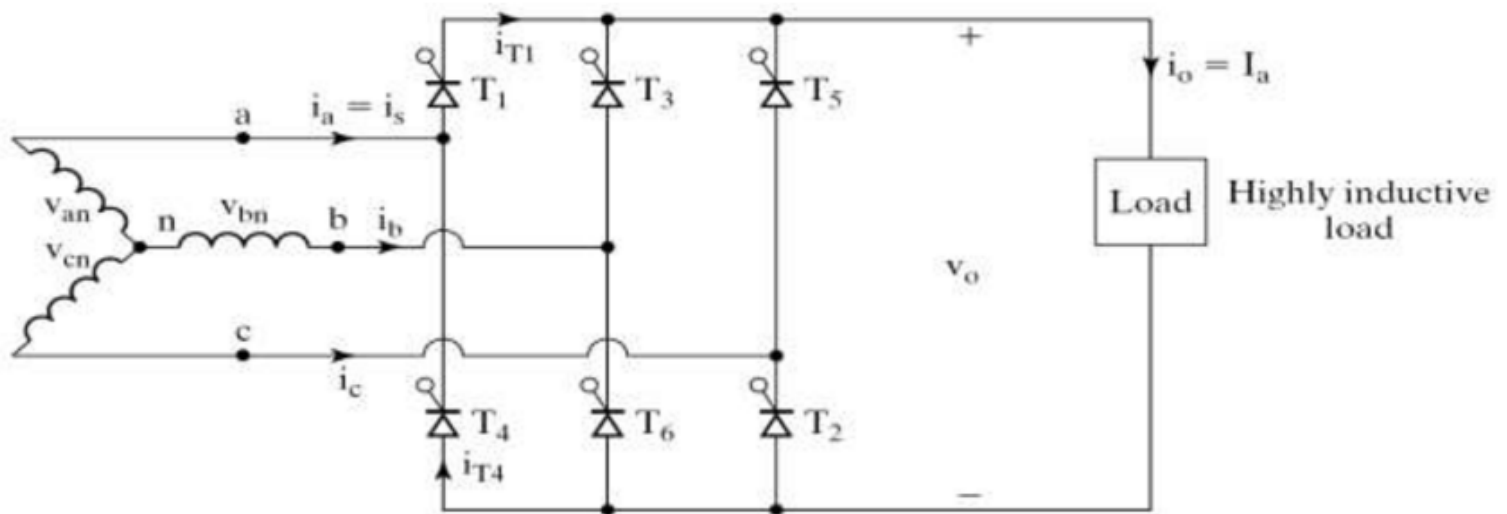


Figure 9.1

5. Experimental Methods

- **Resistive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range ($0-10\text{V}$) to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (9-1).
- 6) Check your circuit connection with the help of your supervisor.
- 7) Turn on the power supply.
- 8) Record the output signal in table (9-1).
- 9) Turn off the power supply.
- 10) Calculate $V_R, V_o, I_R, I_o, RF, P_{dc}, P_{ac}, \eta$, and FF , and record your
- 11) Compare your calculations with the measured values.
- 12) Draw the output wave forms.

Table 9.1 measured values for R load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 9.2 calculated values for R load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and inductive load**

- 1) Construct the circuit diagram with resistive and inductive load.
- 2) Repeat steps (2-8).

Table 9.3 measured values for R-L load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

Table 9.4 calculated values for R-L load:

- **Resistive and inductive with freewheeling diode load**

- 1) Construct the circuit diagram with resistive and inductive load with freewheeling diode.
- 2) Repeat steps (2-8).

Table 9.5 measured values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 9.6 calculated values for R-L load with freewheeling diode

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

- **Resistive and capacitive load**

- 1) Construct the circuit diagram with resistive and capacitive load.
- 2) Repeat steps (2-8).

Table 9.7 measured values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R
0°					
30°					
60°					
90°					
150°					
180°					

Table 9.8 calculated values for R-C load

α	V_{in}	V_0	V_R	I_0	I_R	P_{dc}	p_{ac}	FF	RF	η
0°										
30°										
60°										
90°										
150°										
180°										

6. Conclusions:

Experiment 9

Single phase voltage source inverters

1. Abstract

In this experiment we will construct the voltage source inverter circuit whose purpose is to convert DC voltage to AC voltage.

2. Introduction and Theory:

The principle of operation of this circuit is simply to produce a square wave then filter it for a sinusoidal output, this is done by connecting the load to the DC source via a bridge of GTOs, the sequence of switching of the GTOs will alternately link the load to the positive and then negative ends of the source, then the desired AC output can be obtained by filtering for the low harmonics (theoretically the fundamental harmonic) of the resulting square wave.

3. Used equipments:

- 1-GTOs and Diodes.
- 2- 3 phase AC power supply.
- 3- Circuit breakers.
- 4- R, and L loads.
- 5- Oscilloscope.
- 6- Digital Millimeter.
- 7- Connecting wires.
- 8- Firing Circuit or Pulse Generator.

A) SINGLE PHASE INVERTER

1. SINGLE PHASE HALF-BRIDGE INVERTER:

The circuit in figure (13-1) consists of two transistors. For $T/2$ time interval, if only transistor Q1 starts to conduct, the instantaneous voltage of the terminal of the load is $V_s/2$. In the other $T/2$ time interval, when only transistor Q2 starts to conduct, the instantaneous voltage of the terminal of the load is $-V_s/2$. The system is designed in a way that Q1 and Q2 are not triggered at the same time.

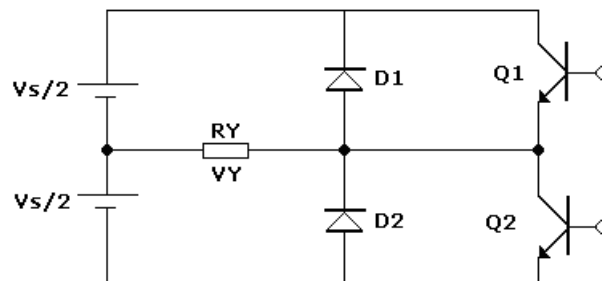


Figure 13.1

Experimental Methods

- **Resistive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (13-1).
- 6) Connect trigger signal to the switches.
- 7) Check your circuit connection with the help of your supervisor.
- 8) Turn on the power supply.
- 9) Record the output signal in table (13-1).
- 10) Turn off the power supply.
- 11) Draw the output wave forms.

Table 13.1 measured value for R load

V_{in}	V_R	I_R

- **Resistive and inductive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (13-1) with R-L load
- 6) Repeat steps (6-11).

Table 13.2 measured value for R-L load

V_{in}	V_R	I_R

2. SINGLE PHASE BRIDGE INVERTER:

Single phase bridge inverter is shown in Figure (13-2). The circuit in the figure consists of four transistors. When transistors Q1 and Q2 start to conduct at the same time, input voltage V_s is observed at the load terminals. When the transistors Q3 and Q4 start to conduct at the same time, reverse of the input voltage, $-V_s$, is observed at the load terminals.

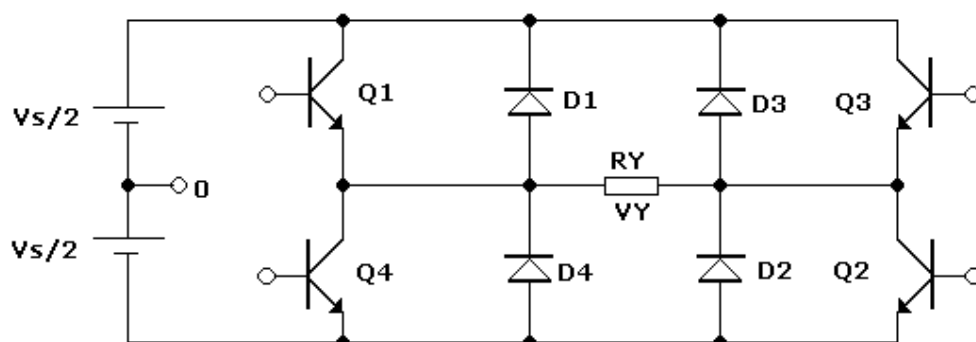


Figure13.2

- **Resistive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0-10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (13-2).
- 6) Connect trigger signal to the switches.
- 7) Check your circuit connection with the help of your supervisor.
- 8) Turn on the power supply.
- 9) Record the output signal in table (13-3).
- 10) Turn off the power supply.
- 11) Draw the output wave forms.

Table 13.3 measured value for R load

V_{in}	V_R	I_R

- **Resistive and inductive load**

- 1) Connect fixed power supply $\pm 15\text{V}$ to trigger unit.
- 2) Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3) Connect control voltage range $(0 - 10\text{V})$ to input U_{st} of the trigger unit.
- 4) Check trigger signal using oscilloscope.
- 5) Turn off the power supply and construct the circuit diagram shown in figure (13-2) with R-L load
- 6) Repeat steps (6-11).

Table 13.4 measured value for R-L load

V_{in}	V_R	I_R

4. Conclusions

Choppers

DC to DC Converters

Abstract:

In this experiment we will construct the Chopper circuit whose purpose is to control DC voltage level, both voltage and current outputs are to be examined.

Introduction and Theory:

The principle of operation of this circuit is based on the fact that the average or DC voltage of a signal is directly proportional to the area enclosed by its waveform.

There are two types of these converters:

- 1- Step-Down (class A) Choppers.
- 2- Step-Up (Class B) Choppers.

These converters uses the GTO as a switching device, the only difference between the GTO and the Thyristor is that the GTO will become OFF as soon as the Gate pulse returns to zero, this means that the Gate pulse period is in fact the period of operation. We will use the GTO to allow the voltage source to be seen by the load for only a part of the operation period, this means that we've changed the area under the waveform, thus the value of the average voltage at the load is changed.

K is called the duty ratio of operation and it's an important feature of the operation mode, it's defined as T_{ON}/T , where T_{ON} is the time through which the GTO is conducting.

Used equipments:

- 1) GTOs and Diodes
- 2) R and RL loads
- 3) Oscilloscope

- 4) Multimeter
- 5) Pulse generation circuit
- 6) 3ph AC voltage source for the pulse generation circuit
- 7) DC voltage supply_

Experimental methods:

1- Class A Chopper with resistive load:

- 1-Connect fixed power supply $\pm 15\text{ V}$ to trigger unit.
- 2- Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3- Connect control voltage range $(0 - 10\text{ V})$ to input U_{st} of the trigger unit.
- 4- Check trigger signal using oscilloscope.
- 5- Turn off the power supply and construct the circuit diagram shown in figure (12-1).
- 6- Connect trigger signal to the switches (IGBTs) .
- 7- Check your circuit connection with the help of your supervisor.
- 8- Turn on the power supply.
- 9- Record the output signal in table (12-1).
- 10- Turn off the power supply.
- 11-Draw the output wave forms.

Circuit diagram:

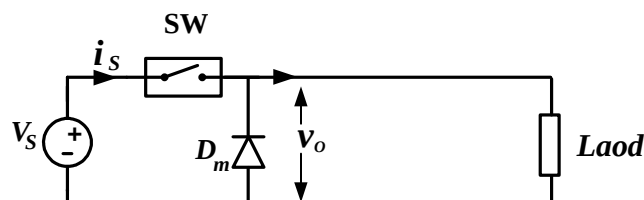


Figure 12-1

Table (12-1)

Duty Ratio	V_{DC}	V_{AC}
$K < 50\%$		
$K = 50\%$		
$K > 50\%$		

2- Class A Chopper with RL load:

- 1- Construct the circuit with RL load.
- 2- Repeat the steps (7-11).

3- Class B Chopper with resistive load:

- 1-Connect fixed power supply $\pm 15\text{ V}$ to trigger unit.
- 2- Connect line voltage L_1, L_2, L_3 to trigger unit.
- 3- Connect control voltage range $(0 - 10\text{ V})$ to input U_{st} of the trigger unit.
- 4- Check trigger signal using oscilloscope.
- 5- Turn off the power supply and construct the circuit diagram shown in figure (12-2).
- 6- Connect trigger signal to the switches (IGBTs) .
- 7- Check your circuit connection with the help of your supervisor.
- 8- Turn on the power supply.
- 9- Record the output signal in table (12-2).
- 10- Turn off the power supply.
- 11-Draw the output wave forms.

Circuit diagram:

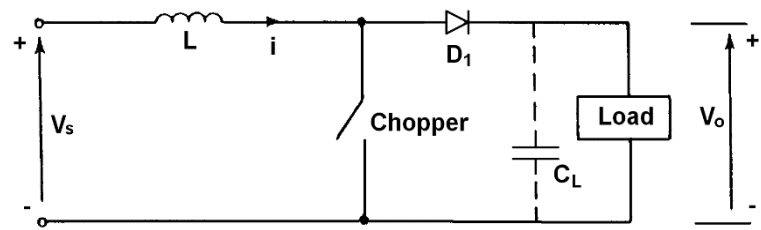


Figure 12-2

Table (12-2)

Duty Ratio	V_{DC}	V_{AC}
$K < 50\%$		
$K = 50\%$		
$K > 50\%$		

4- Class B Chopper with RL load:

- 1- Construct the circuit with RL load.
- 2- Repeat the steps (7-11).

